

Digital Invitation and Attendance Management System

Complete System Features and System Requirements

1. System Overview

The **Digital Invitation and Attendance Management System for University Graduation Ceremonies** is a web and mobile-based platform designed to improve the organization and management of graduation events.

The system replaces traditional paper-based invitation methods with a **secure digital system using QR codes**, allowing universities to manage invitations, monitor attendance in real time, control guest access, and organize seating arrangements efficiently.

The system integrates **web and mobile technologies**, enabling administrators to manage events through a web dashboard while graduates and guests access invitations through mobile devices.

2. System Architecture

The system follows a **three-tier architecture**:

1. Presentation Layer

User interfaces used by system users.

Includes:

- Web Admin Dashboard
- Mobile Application
- QR Code Scanner Interface

Technologies:

- **React.js (Web Interface)**
- **Flutter (Mobile Application)**

2. Application Layer

Handles system logic and processing.

Functions:

- Invitation generation
- QR code generation

- Attendance verification
- Seat allocation
- Notification services
- Role management

Technology:

- **Node.js with Express.js**

3. Data Layer

Stores and manages system data.

Includes:

- User information
- Invitations
- Attendance records
- Seating data
- Event data

Technology:

- **MongoDB Database**

3. Technology Stack

The system will be developed using modern web and mobile technologies.

Component	Technology
Frontend (Web)	React.js
Mobile Application	Flutter
Backend	Node.js + Express.js
Database	MongoDB
Authentication	JWT (JSON Web Token)

QR Code Generation QR Code Libraries

Component	Technology
Encryption	AES Encryption
Notifications	SMS / Email API
Hosting	Cloud Server (AWS / DigitalOcean)

4. System Users

The system includes four main user roles:

1. Administrator

University management responsible for organizing the ceremony.

Responsibilities:

- Create graduation events
- Register graduates
- Manage invitations
- Monitor attendance
- Generate reports

2. Graduate

Students graduating from the university.

Permissions:

- Receive digital invitation
- Invite one guest
- Share guest QR code
- View ceremony information

3. Guest

A person invited by a graduate.

Permissions:

- Receive invitation
- Use QR code for entry
- Access seating information

4. Special Guest (VIP)

Guests invited directly by the university.

Examples:

- Ministers
- Government officials
- Sponsors
- University partners

Permissions:

- Special QR code
- Priority entry
- VIP seating area

5. System Functional Features

5.1 User Role Management

The system supports role-based access control.

Functions:

- Admin user management
- Graduate registration
- Guest linking to graduate
- Special guest registration

Benefits:

- Controlled access
- Clear responsibilities
- Secure system operations

5.2 Invitation Management

The system generates and manages digital invitations.

Features:

- Personalized invitations
- QR code generation
- Email/SMS invitation delivery
- Invitation tracking

Benefits:

- Paperless invitations
- Faster distribution
- Secure verification

5.3 Invitation Limitation Policy

The system strictly limits invitations to avoid overcrowding.

Rules:

- One invitation for each graduate
- One invitation for one guest
- Maximum two invitations per graduate

Admin can override in special situations.

Benefits:

- Fair invitation distribution
- Crowd control
- Easy event planning

5.4 QR Code-Based Attendance System

Each invitation contains a **unique QR code**.

Features:

- QR code scanning at entry gates

- One-time validation
- Duplicate scan prevention
- Real-time attendance recording

Benefits:

- Fast entry process
- Accurate attendance records
- Improved security

5.5 Special Guest Management

The system supports special handling for VIP guests.

Features:

- Manual VIP registration
- Dedicated invitation templates
- Separate QR code category
- Priority entry gate

Benefits:

- Professional guest handling
- Better ceremony organization

5.6 Intelligent Seat Allocation

The system automatically assigns seats using structured logic.

Seating hierarchy:

Faculty → Department → Graduate → Guest

Categories:

- Faculty seating zones
- Department sections
- Graduate seats
- Guest seats

- VIP seating area

Benefits:

- Organized seating
- Efficient space usage

5.7 Performance-Based Seating

The system ranks graduates according to academic performance.

Process:

1. System reads GPA/percentage
2. Ranks graduates
3. Assigns seats automatically

Higher performing students receive priority seats.

Benefits:

- Fair and transparent seating
- Recognition of academic achievement

5.8 Degree-Level Priority Seating

Graduates are grouped by degree level.

Priority order:

1. PhD
2. Master's
3. Bachelor's

Within each degree level, seating follows **performance ranking**.

Benefits:

- Reflects academic hierarchy
- Maintains ceremony professionalism

5.9 Seat Position Validation

Each QR code contains seat information.

When scanned:

- System verifies assigned seat
- Detects incorrect seating
- Alerts administrators

Benefits:

- Prevents seat misuse
- Maintains order during ceremony

5.10 Entry Gate Management

Separate entry gates improve event control.

Gate types:

- Graduate and Guest gate
- Special Guest (VIP) gate

Benefits:

- Faster entry
- Reduced congestion

5.11 Real-Time Notification System

The system sends notifications to users.

Examples:

- Ceremony reminders
- Entry instructions
- Seat location alerts
- Admin alerts

Communication methods:

- SMS
- Email
- Mobile notifications

5.12 Dashboard and Live Monitoring

Administrators can monitor events through a dashboard.

Displays:

- Total attendees
- Graduate attendance
- Guest attendance
- Special guest attendance
- Faculty-based statistics

Benefits:

- Real-time event management
- Immediate issue detection

5.13 Reporting and Audit Logs

The system generates reports for analysis.

Reports include:

- Attendance reports
- Invitation usage
- Guest participation
- VIP attendance

Reports can be exported to:

- PDF
- Excel
- CSV

Benefits:

- Transparency
- Record keeping
- Post-event analysis

5.14 Security and Access Control

The system implements strong security mechanisms.

Security features:

- Encrypted QR codes
- AES data encryption
- Role-based access control
- Secure authentication
- Duplicate scan prevention

Benefits:

- Prevents unauthorized access
- Protects event integrity

5.15 Sustainability and Cost Efficiency

The system reduces environmental impact.

Benefits:

- Eliminates paper invitations
- Reduces printing costs
- Eco-friendly solution
- Supports digital transformation

6. Functional Requirements

The system must be able to:

1. Allow administrators to create graduation events.
2. Register graduates and assign departments.
3. Generate digital invitations with QR codes.
4. Allow graduates to invite one guest.
5. Register special guests.
6. Scan QR codes to verify attendance.

7. Automatically assign seats.
8. Validate seat positions using QR codes.
9. Provide real-time attendance monitoring.
10. Generate attendance reports.

7. Non-Functional Requirements

Performance

- System should process QR scans in **less than 2 seconds**.

Security

- Data must be protected using **AES encryption**.

Scalability

- System should support **thousands of attendees**.

Usability

- Interfaces must be simple and user-friendly.

Reliability

- System should operate continuously during events.

Availability

- System should be accessible via **web and mobile devices**.

8. Database Requirements (MongoDB Collections)

Main collections:

- Users
- Graduates
- Guests
- SpecialGuests
- Events
- Invitations
- QRcodes

- AttendanceLogs
- Seats
- Departments
- Faculties

9. Summary

The proposed Digital Invitation and Attendance Management System integrates **web and mobile technologies** to improve graduation ceremony management. The system enhances efficiency through **QR code attendance verification, intelligent seating allocation, real-time monitoring, and secure digital invitations**. By replacing traditional paper-based methods, the system provides a **modern, scalable, and secure solution** for university graduation events.